

The strong negative influence of WBC and heart rate outweigh the positive influence of respiratory rate, leading to an overall low risk of sepsis (−4.633).

Table 3: XAI significance of waterfall plot

XAI aspect	Explanation with waterfall plot
Transparency	We see exactly which features drove the decision and in which direction.
Local interpretability	Tailored to this specific patient — not a general trend.
Consistency	Matches domain knowledge (e.g., very low WBC decreases risk).
Trust	Clinicians can understand the rationale — increasing confidence in model output.
Auditability	Useful in regulated environments to defend decisions on individual predictions.

This plot provides a global view of feature importance across all patients in the test set. Each dot = one patient.

Feature	Role	SHAP trend
resp_rate	Most important overall	High values → higher sepsis risk
wbc	Strongly bi-directional	Low values → lower risk
heart_rate	Important signal	High values → higher risk
lactate	Moderate influence	Non-linear behaviour
temp, age	Minimal contribution	Weak or neutral

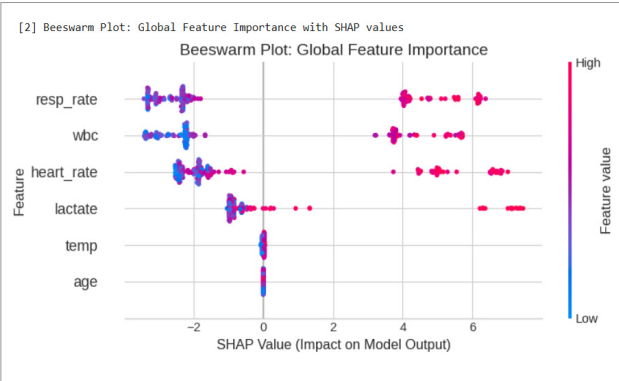


Figure 5: Beeswarm plot

Beeswarm plot: Now let us see what the beeswarm plot is showing.

- Each dot = one patient in the dataset.
- Y-axis = features used by the model.
- X-axis = SHAP value (how much a feature pushed prediction up or down).

Feature	Insight
resp_rate	Most important feature. High values consistently push predictions strongly up → higher sepsis risk.
wbc	Bidirectional impact. High WBC pushes risk up (infection likely). Low WBC pushes risk down.
heart_rate	Important signal. High heart rate = elevated sepsis risk.
lactate	Moderate influence. Shows non-linear behaviour—some high values impact prediction strongly, others don't.
temp	Weak impact. Most dots cluster around 0 SHAP value.
age	Minimal contribution overall. Likely not a key predictor in this model for sepsis.

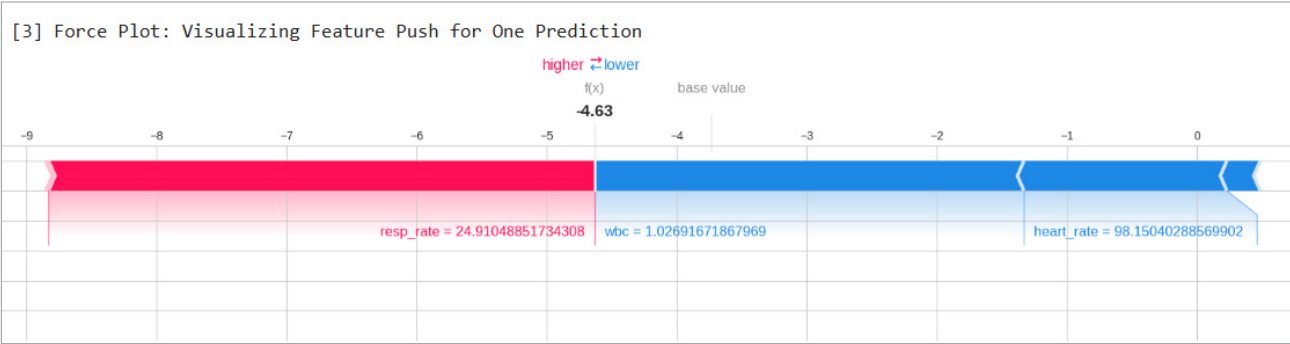


Figure 6: Force plot